

Vertical Narrative in Virtual Reality: A Rhythm-Embodiment Design Framework (REDF)

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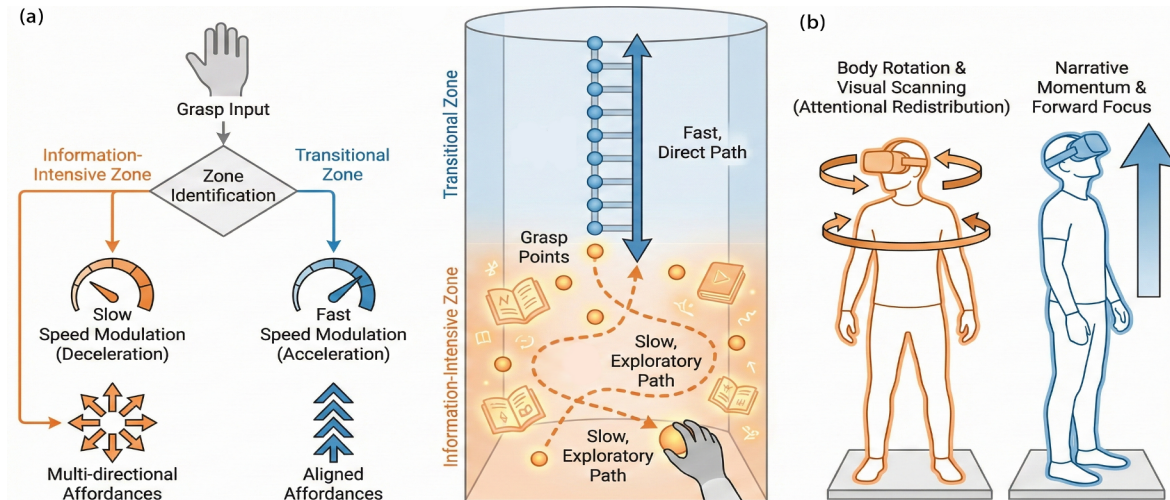


Figure 1: Rhythm-Embodiment Design Framework (REDF): (a) Zone-based modulation of locomotion speed and grasp-point configuration; (b) Embodied attention patterns across transitional and information-intensive vertical zones.

ABSTRACT

VR narratives often prioritize horizontal exploration, leaving vertical experience underexplored. We propose REDF, a Rhythm-Embodiment design framework that treats rhythm as a spatial, embodied experience shaped by users' movement and the arrangement of graspable elements. REDF supports vertical narrative pacing via (1) post-grasp locomotion-gain modulation for implicit time dilation in information-intensive zones and (2) multi-directional grasp points that encourage body rotation and visual scanning. In a between-subject study ($N=15$), information-intensive zones showed longer engagement and higher IPQ presence ratings, supporting REDF as a practical approach for designing vertical narrative experiences in VR.

Index Terms: Virtual Reality, Vertical Narrative, Embodied Interaction, Presence, Immersive Storytelling

1 INTRODUCTION

A horizontal bias persistently underlies narrative design in virtual reality. Yet the narrative potential of the vertical dimension remains underexplored. Multiple studies have shown that vertical experiences in VR can induce significant physiological and psychological effects. For example, exposure to height can trigger fear and excitement responses similar to those in real situations [2], and vertical locomotion has been demonstrated to create spatial perception and immersion distinctly different from horizontal movement [4]. AI-

though existing studies reveal important perceptual characteristics of vertical space, they largely focus on isolated physiological or spatial effects. There remains a lack of systematic design frameworks that integrate the static spatial structure of vertical environments with users' dynamic bodily interactions to support narrative experience in virtual reality. In this work, rhythm is understood not as a purely temporal construct, but as a spatially situated and embodied experience. It emerges from the interplay between users' bodily movement and the static spatial arrangement of graspable elements, shaping the alternation of tension and release across vertical space. To address this, this paper proposes a Rhythm-Embodiment design framework, through which we specifically design and implement a vertical locomotion mechanism based on grasping interaction. This transforms the user's vertical movement from simple spatial displacement into a narrative process modulated by the body.

2 RHYTHM-EMBODIMENT VERTICAL NARRATIVE MECHANISM

2.1 Locomotion Speed Modulation in Information-Intensive Zones

To support narrative comprehension and sustained presence, we modulate post-grasp locomotion speed based on the narrative information load. In Information-Intensive Zones, grasp-based locomotion is intentionally decelerated (Fig. 1a). This controlled deceleration introduces a "time-dilation" effect, operationalized as reduced locomotion gain following grasp actions, allowing users to remain longer within narratively dense environments without explicitly interrupting the experience.

This mechanism slows progression in a manner that feels physically grounded. Instead of pausing the scene, time is expanded through bodily effort and motion resistance, aligning with the concept of place presence maintained through sensorimotor contingencies [8]. By manipulating locomotion gain—specifically the ratio

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between the user's physical pulling motion and the resulting virtual displacement—the system implicitly regulates the pace of exploration by altering sensorimotor contingencies that underlie perception and presence in immersive virtual environments [3].



Figure 2: (left) Transitional zone with aligned grasp points to maintain upward flow; (right) Information-intensive zone with multi-directional grasp points encouraging rotation and visual scanning.

This implicit modulation allows exploration pace to better align with the narrative information density of the environment. As illustrated in (Fig. 2), this pacing strategy is implemented through distinct grasp-point configurations and locomotion behaviors across transitional and information-intensive vertical zones.

2.2 Directional Distribution of Grasping Points and Embodied Attention

In addition to temporal modulation, we regulate embodied attention through the directional distribution of grasping points. In information-intensive zones, grasping points are distributed across multiple directions rather than a single upward axis (Fig. 1a). This requires users to rotate their bodies and shift gaze laterally while progressing vertically (Fig. 1b).

Such multi-directional grasping transforms vertical ascent into an embodied exploratory activity. Compared to Transitional Zones (Fig. 2), Information-Intensive Zones prompt active scanning and reorientation. This functions as embodied guidance [7], where physical reorientation redistributes visual attention across narrative elements. This design supports experiential presence by coupling narrative engagement with bodily movement, reinforcing the perception that the user's body is meaningfully situated within the virtual space rather than merely traversing it. During on-site experiment, the frequent torso rotations and lateral head movements observed in participants provide behavioral evidence of this active spatial situatedness.

2.3 Coordinating Rhythm and Embodiment through Grasp-Based Interaction

REDF achieves implicit narrative pacing by manipulating physical effort rather than relying on scripted triggers. In Transitional Zones, directionally aligned grasping points and faster locomotion reinforce narrative momentum and forward focus (Fig. 1b), maintaining continuity between segments.

The integration of these configurations frames vertical movement as an experiential narrative process enacted through the body, echoing the principles of embodied storytelling, where narrative engagement is facilitated through the physical effort and pacing of movement [5]. By embedding pacing within grasp-based interaction, the mechanism enhances presence while preserving immersion. As participants interact with the system, vertical movement becomes an expressive narrative component, where rhythm and meaning emerge through embodied action [1].

2.4 Evaluation

We recruited 16 participants, one withdrew due to a headset shutdown mid-session, leaving $N=15$ for a between-subject comparison ($n=8$ Information-Intensive vs. $n=7$ Transitional). Dwell time and revisits were extracted from interaction logs. Presence was assessed using the Igroup Presence Questionnaire (IPQ) [6] on a 7-point Likert scale, centered to $[-3,3]$ (0=neutral); reverse-worded items were sign-inverted before computing subscale means. We report Spatial Presence (SP), Involvement (INV), and Experienced Realism (REAL). Narrative comprehension was assessed via a brief post-experience interview. Information-Intensive Zones showed longer engagement (revisits: $M=1.71$ vs. 0.14 ; dwell time: $M=46.82s$ vs. $21.35s$) and higher IPQ ratings (SP: 1.03 ± 0.77 vs. 0.11 ± 0.67 ; INV: 0.91 ± 0.57 vs. 0.04 ± 0.82 ; REAL: 0.34 ± 0.58 vs. 0.07 ± 0.81). Interviews supported narrative comprehension, with more participants in the Information-Intensive condition identifying the growth theme (7/8 vs. 3/7).

3 CONCLUSION

This work explores the narrative potential of vertical space in VR by proposing a Rhythm–Embodiment design framework (REDF) and implementing a grasp-based vertical locomotion mechanism. By modulating locomotion speed and grasp-point orientation across spatial zones, REDF turns vertical movement into a body-modulated narrative process. A lightweight between-subject evaluation ($N=15$), combining dwell behavior, IPQ presence ratings, and brief interviews, suggests that Information-Intensive Zones promote longer engagement, higher perceived presence, and clearer narrative comprehension. To facilitate practical use, we also developed a one-click Unreal Engine layout tool for REDF-aligned spatial design. Future work will expand the evaluation and apply the framework to more complex narrative structures.

ACKNOWLEDGMENTS

This work is supported by National Social Science Fund Major Program, under Grants No.25ZD07.

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